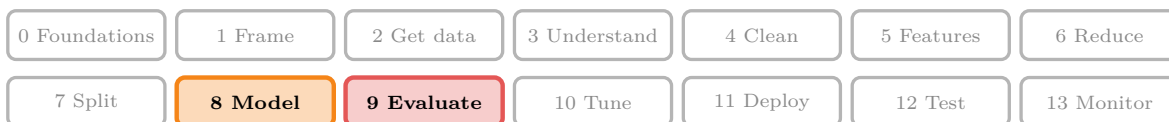


Bagging: Bootstrap Aggregation

Contents

1. Overview	2
2. The core idea	2
2.1 Bootstrapping	3
2.2 Aggregation	4
2.3 Drawing with replacement	4
3. Why bagging works	6
3.1 The bias-variance problem	6
3.2 How bagging lowers variance	6
4. When to use bagging	7
5. Bagging by hand	8
5.1 The data	8
5.2 Bootstrapping: three samples, three trees	8
5.3 Aggregation: the vote	9
6. Types of bagging	9
6.1 Bagging	10
6.2 Pasting	10
6.3 Random subspaces	10
6.4 Random patches	10
7. Summary	11
8. Sources	11
9. Key terms	11

Where this fits: Pipeline map steps 8 (Model), 9 (Evaluate). See the [Course map](#).



- **Builds on:** Bias-variance trade-off ([Note 62](#)); Overfitting ([Note 91](#)); Grid and random search ([Note 99](#)); Decision trees ([Note 100](#)); Ensemble learning ([Note 101](#)).
- **Leads to:** Random forest ([Note 108](#)).
- **Compare with:** Boosting ([Note 101](#)); Voting ensembles ([Note 104](#)); Cross-validation ([Note 104](#)); Random forest ([Note 108](#)).

1. Overview

Key point: Bagging trains many copies of one algorithm, each on a different random sample of the observations (bootstrapping), and combines their predictions by vote or mean (aggregation). Bagging keeps the low bias of a flexible model and cuts its variance.

Bagging, short for **bootstrap aggregation**, is one of the two most important ensemble techniques, with boosting (the [introduction to ensemble learning Note](#), section 4.3, introduced it). The name joins its two steps: **B**ootstrapping and **AGG**regation.

This Note covers the core idea, why bagging works, when to use it, a small worked example in code, and the four ways to sample: bagging, pasting, random subspaces and random patches. The next two Notes apply it to classification ([bagging classifier](#)) and regression ([bagging regressor](#)).

The Notebook (`notebook.ipynb`) runs the worked example.

2. The core idea

Key point: Same algorithm for every base model; different random observations for each; then vote or average.

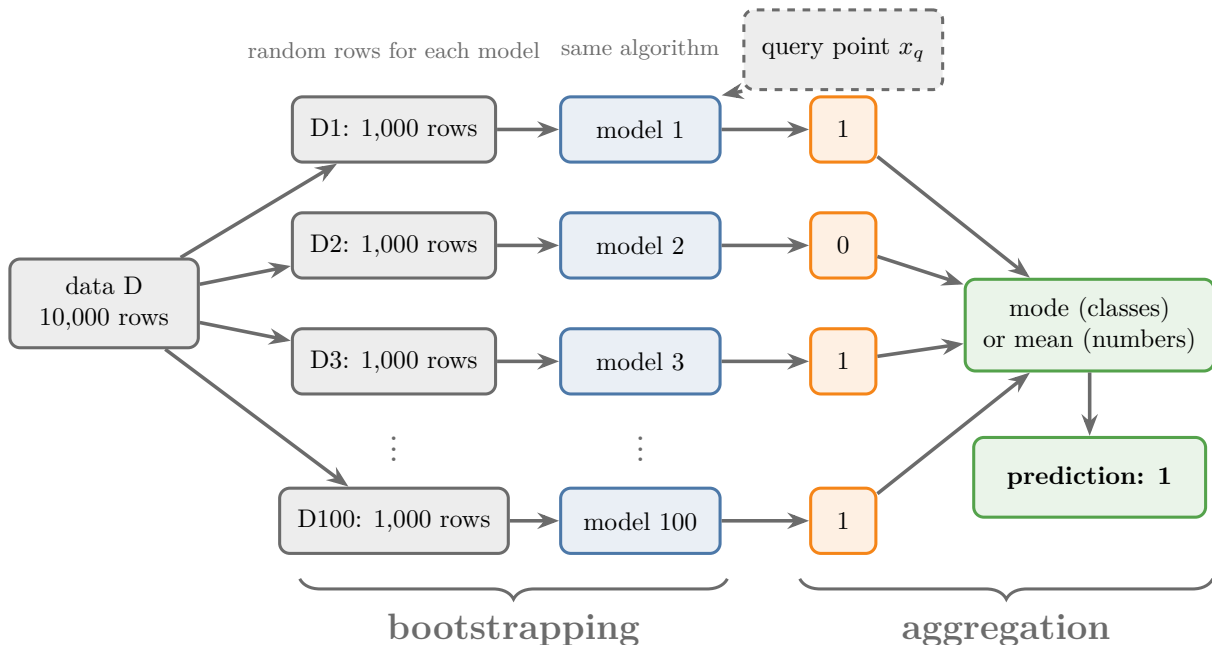


Figure 1: Bagging: each base model trains on its own random sample of the observations; the predictions are combined by mode or mean

Take a classification dataset **D** of 10,000 **observations** (records; each is one row of the data table) with a yes/no **target** (the output we predict). Figure 1 shows the two steps.

2.1 Bootstrapping

Key point: Draw a random sample of observations for each base model and train it on that sample.

1. Choose a number of base models, say 100 (it could be 10 or 1,000). They all use the **same algorithm**: all decision trees, all KNN, or all SVMs. Unlike voting, bagging does not mix algorithms: by definition it makes many versions of one predictor (Breiman, 1996, section 1).
2. Choose how many observations each model sees, say 1,000.
3. Draw 1,000 observations at random from **D**: this is dataset **D1**. Train model 1 on **D1**.
4. Draw another random 1,000 observations: **D2**, for model 2. **D2** differs from **D1**, because the observations are random. And so on, up to **D100**.

In bagging, drawing these random samples is called **bootstrapping**, after the statistics technique of resampling a dataset with replacement (section 2.3; ESL §8.7). Each base model is trained on a different subset, so each learns something slightly different, and the ensemble gets its variety from the data rather than from the algorithms.

2.2 Aggregation

Key point: Send the query point to every model and combine their answers: the mode for classification, the mean for regression.

For a new query point x_q , every trained model predicts, and the answers are combined as in every ensemble (the [ensemble learning Note](#), section 3): the most common answer, the **mode**, for classification (in Figure 1, most models say 1, so the answer is 1) and the mean for regression. This combining step is the **aggregation**.

So the only difference from voting (the [voting ensemble Note](#)) is where the variety comes from:

	Voting	Bagging
Base models	different algorithms	one algorithm
Data each model sees	the same data	a different random sample

Because the base models differ, the probability argument of the voting ensemble Note applies, and the combined model usually performs better than its base models. In practice scikit-learn does all of this for us; the [bagging classifier Note](#) shows how.

2.3 Drawing with replacement

Key point: In a bootstrap sample an observation can be drawn more than once, and some are never drawn: on average a sample holds about 63% of the distinct observations.

Observations can be drawn **with replacement**: after an observation is drawn, it is put back and can be drawn again. Picture a bag of numbered balls: we pull one out, write down its number, and drop it back in before the next pull. Figure 2 draws 10 observations from 10, with replacement, twice.

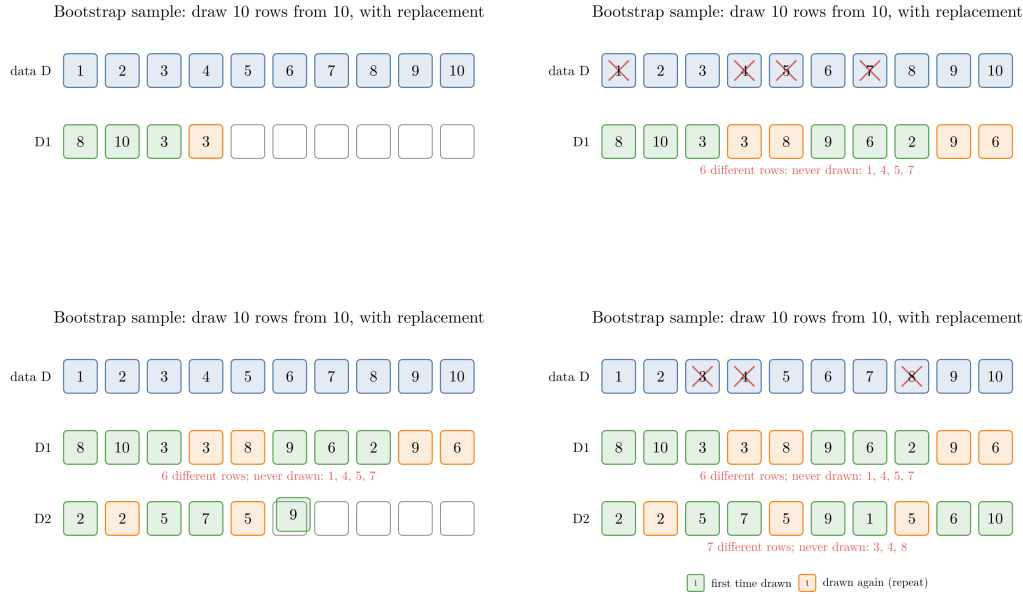


Figure 2: Drawing two bootstrap samples of 10 observations from 10: green rows are drawn for the first time, orange rows are repeats, crossed rows are never drawn

- D1 holds only 6 different observations: 3, 6, 8 and 9 appear twice, and observations 1, 4, 5 and 7 are never drawn.
- D2 holds 7 different observations: 2 appears twice and 5 three times, while 3, 4 and 8 are missing.

A sample of the same size as the data, drawn with replacement, is a **bootstrap sample** (Efron and Tibshirani, 1993; the term was used in the [Ridge key points Note](#)). Drawing observations without replacement is also possible; section 5 calls that pasting.

Extra: How many different observations does a bootstrap sample hold?

1. **In words:** one draw misses a given observation with probability $1 - 1/n$. All n draws miss it with probability $(1 - 1/n)^n$. So the share of observations that appear at least once is $1 - (1 - 1/n)^n$.
2. **Formula:** as n grows, $(1 - 1/n)^n$ approaches $1/e$, so

$$\text{share of distinct observations} = 1 - \left(1 - \frac{1}{n}\right)^n \longrightarrow 1 - \frac{1}{e} \approx 0.632$$

3. **Example:** for $n = 10$: $(0.9)^{10} = 0.349$, so $1 - 0.349 = 0.651$, about 6.5 distinct observations out of 10 (Figure 2 got 6 and 7). For $n = 10,000$ the share is 0.632. The Notebook checks this by simulation: 0.655 for $n = 10$, 0.632 for $n = 1,000$ and above. The roughly 37% of observations a model never sees are its **out-of-bag** observations; the [bagging classifier Note](#) uses them to score the model.

3. Why bagging works

Key point: Bagging takes low-bias, high-variance models, such as fully grown trees, and averages away much of their variance, leaving a model with low bias and low variance.

3.1 The bias-variance problem

Key point: We want low bias and low variance, but in a single model reducing one usually raises the other.

From the [bias-variance Note](#): **bias** is a model's inability to fit even its training data, and **variance** is how much its predictions change when the training data changes. A good model has low bias (accurate on the training data) and low variance (consistent results when the data changes slightly).

The trouble is that the two pull against each other. Most algorithms end up either low bias, high variance (they overfit) or high bias, low variance (they underfit).

3.2 How bagging lowers variance

Key point: A change in the data is spread across many models, so no single change can swing the ensemble.

Bagging uses base models that are **low bias, high variance**:

- a **fully grown decision tree** (`max_depth=None`, the [decision tree hyperparameters Note](#)), which fits its training data perfectly but overfits;
- other **unstable** models, whose fit changes a lot when the data changes a little: Breiman (1996, section 1) found trees and neural networks unstable, and nearest-neighbour methods stable.

Their bias is already low, and averaging leaves it about where it was; what averaging cuts is the variance (ESL §8.7, §15.2; Breiman, 1996, section 4).

Suppose we replace 100 of the 10,000 observations with very different, noisy ones. A single tree trained on all the data would change its logic completely because of those 100 observations. In bagging, each tree sees a random 5,000 observations, so the 100 new ones are spread out: one tree gets 10 of them, another 50, another none, another 35.

No single tree absorbs all the change, so the behaviour of each tree changes less, and the average changes even less. The ensemble gives consistent results: **low variance**. Since its base models were already accurate (low bias), we end up with the combination we wanted: **low bias, low variance**.

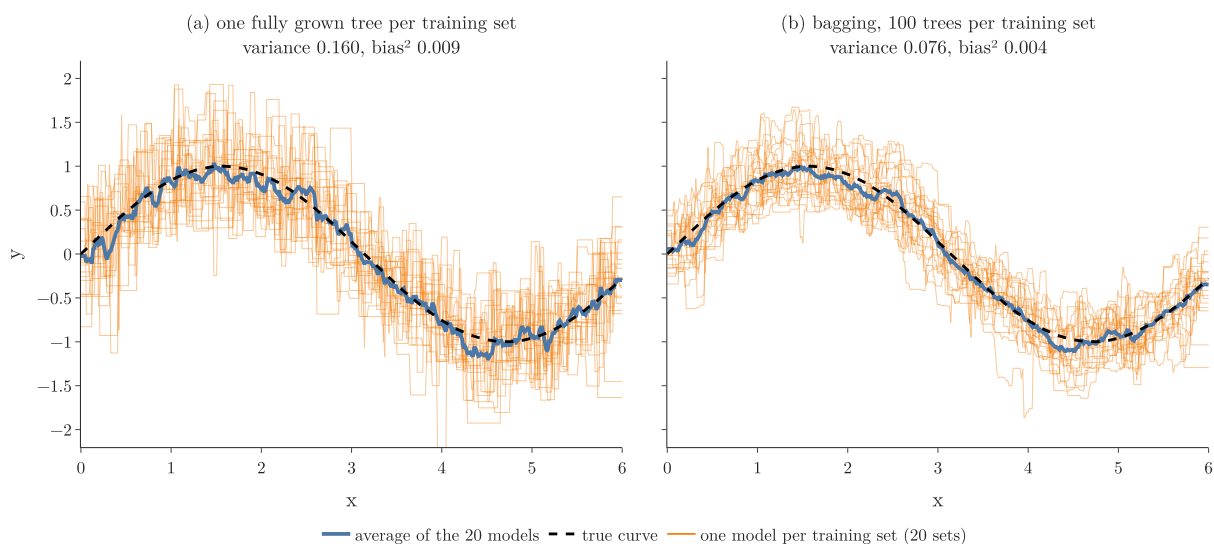


Figure 3: The same experiment repeated on 20 different training sets: (a) a single fully grown tree, (b) bagging with 100 trees. The spread of the orange curves is the variance

Figure 3 shows this. We draw 20 different training sets of 60 noisy points from a sine curve and fit a model to each:

- **(a) one fully grown tree per set:** the 20 curves swing wildly around the truth. Average variance **0.160**.
- **(b) bagging with 100 trees per set:** the 20 curves stay close together. Average variance **0.076**, less than half.

The squared bias stays tiny in both (0.009 and 0.004): both average curves follow the true curve.

4. When to use bagging

Key point: Whenever the model is low bias and high variance (it overfits), try bagging; it is not limited to decision trees.

Bagging is worth trying whenever a model overfits: low bias, high variance. Breiman (1996, section 1) calls such models **unstable**: a small change in the training data causes a large change in the model. Bagging helps unstable models and can slightly hurt stable ones, such as nearest-neighbour methods (Breiman, 1996, sections 1 and 6.3). Bagging is also what makes **random forests** (bagging with decision trees, see the [random forest Note](#)) so popular.

A common misunderstanding is that bagging only works with decision trees. Trees are the usual choice because they are unstable, so they gain the most. **Any algorithm** can be bagged, but the gain depends on how unstable it is: the [bagging classifier Note](#) bags KNN and SVMs too, and they gain little or nothing.

5. Bagging by hand

Key point: Three trees, each trained on 8 observations drawn with replacement, then a majority vote.

To see each step, we bag three decision trees on a tiny dataset.

5.1 The data

Key point: Versicolor against virginica, with sepal width and petal length; 10 training observations.

From the iris data (the [softmax regression Note](#)) we keep versicolor (class 1) and virginica (class 2), two species that overlap, and two **features** (input variables, columns of the data table): sepal width and petal length. That leaves 100 observations.

The rows of the table are stored species by species, so we first shuffle them with `df.sample(100)`. Then we take tiny sets so that every step can be printed: 10 random observations for training, 5 for validation and 5 for testing.

5.2 Bootstrapping: three samples, three trees

Key point: Each tree gets its own 8 observations drawn with replacement, and grows fully on them.

Python: One bootstrap sample and its tree.

```
# 8 rows from the 10, with replacement
df_bag = df_train.sample(8, replace=True)
X, y = df_bag.iloc[:, 0:2], df_bag.iloc[:, -1]
# max_depth not set: a fully grown tree
tree = DecisionTreeClassifier().fit(X, y)
sample(n) draws n random rows; replace=True puts each row back after drawing it, so
rows can repeat. The default, replace=False, never repeats a row.
```

Running this three times gives three different samples. In the Notebook, the first sample holds row 91 three times and row 69 twice; the third holds row 105 three times. With so few observations, each tree makes a single split on petal length, but each at a different place:

Tree	Split	Validation accuracy (5 observations)
1	petal length $\leq 5.20 \rightarrow$ class 1	0.4
2	petal length $\leq 4.90 \rightarrow$ class 1	0.8
3	petal length $\leq 4.95 \rightarrow$ class 1	0.6

5.3 Aggregation: the vote

Key point: For a virginica with sepal width 2.2 and petal length 5.0, the trees say 1, 2, 2; the majority says 2, which is right.

We send a new flower to all three trees: sepal width 2.2, petal length 5.0. The flower is a virginica (class 2).

- Tree 1 (split at 5.20): 5.0 is below, so **class 1**.
- Tree 2 (split at 4.90): above, so **class 2**.
- Tree 3 (split at 4.95): above, so **class 2**.

The majority says **2**, which is correct, although tree 1 alone would have been wrong. Each tree here saw only 8 observations, so it made just one split. With more observations, trees grow deeper: in the [random forest Note](#), section 5.3, trees trained on 100 observations reach depths 3 to 6.

6. Types of bagging

Key point: Sample observations with replacement (bagging) or without (pasting), sample features only (random subspaces), or both (random patches).

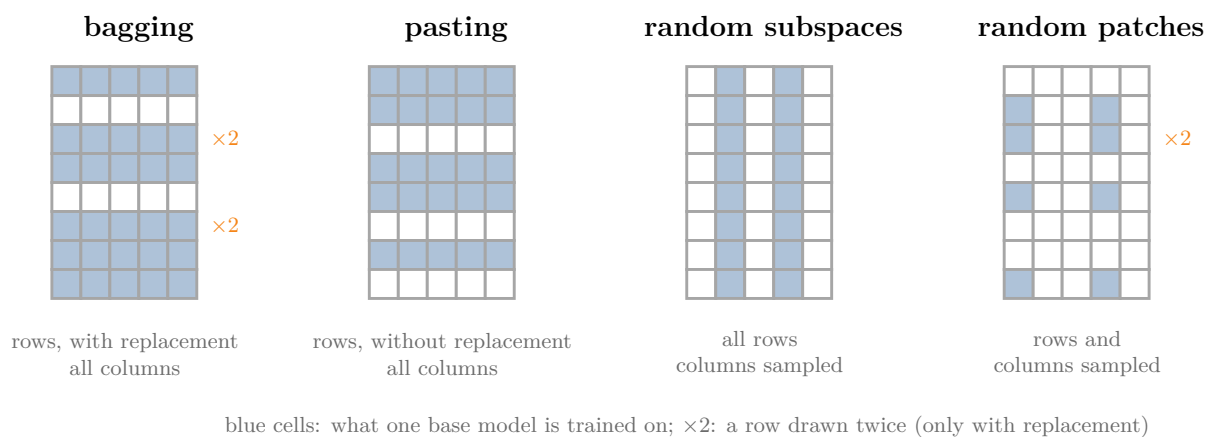


Figure 4: What one base model sees under each type: blue cells are the rows (observations) and columns (features) it is trained on

So far we sampled **observations** (rows of the table). We can sample **features** (columns) too: instead of giving each tree every feature, we give it a random subset. Feature sampling is meant for data with many features; the [bagging classifier Note](#), section 2.4, shows it hurting when there are only two. Combining these choices gives four types (Figure 4), each with its own name in the literature (scikit-learn User Guide, “Bagging meta-estimator”).

6.1 Bagging

Key point: Observation sampling with replacement.

What we have done so far: each model gets random observations, drawn with replacement, and all the features (Breiman, 1996).

6.2 Pasting

Key point: Observation sampling without replacement: no observation repeats inside a sample.

Pasting (Breiman, 1999) is the same as bagging except that observations are drawn **without replacement**. In pandas, `df_train.sample(8)` gives 8 rows with no repeats, because `replace=False` is the default.

6.3 Random subspaces

Key point: Feature sampling only: every model gets all the observations but only some of the features.

In **random subspaces** (Ho, 1998), there is no observation sampling. Each model gets every observation but a random subset of the features, drawn with or without replacement.

Python: Sampling columns.

```
df1.sample(2, replace=True, axis=1)    # 2 random columns
```

`axis=1` tells `sample` to pick columns instead of rows. With `replace=True` the same column can be picked twice.

6.4 Random patches

Key point: Observation sampling and feature sampling together.

In **random patches** (Louppe and Geurts, 2012), each model gets a random set of observations **and** a random set of features. In pandas we chain two samples:

Python: Sampling rows, then columns.

```
df1.sample(8, replace=True).sample(2, replace=True, axis=1)
```

All four aim at the same thing: give each base model different data, so the models differ. The [bagging classifier Note](#) runs all four in scikit-learn and compares them.

7. Summary

Type	Observations (rows)	Features (columns)
Bagging	sampled, with replacement	all
Pasting	sampled, without replacement	all
Random subspaces	all	sampled (with or without replacement)
Random patches	sampled	sampled

- Bagging = bootstrapping (each model trains on random observations) + aggregation (mode or mean of the predictions).
- All base models use the same algorithm; the variety comes from the data.
- A bootstrap sample holds about 63.2% of the distinct observations; the rest are out-of-bag.
- Bagging keeps the low bias of a flexible model and cuts its variance: 0.160 to 0.076 in Figure 3.
- Use it whenever a model overfits (an unstable model); decision trees are the usual choice, but any unstable algorithm can be bagged.

8. Sources

- Breiman, L. (1996). “Bagging Predictors”. *Machine Learning* 24(2), 123–140.
- Breiman, L. (1999). “Pasting Small Votes for Classification in Large Databases and On-Line”. *Machine Learning* 36(1), 85–103.
- Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman and Hall.
- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning* (ESL), 2nd ed. Springer, sections 8.7 and 15.2.
- Ho, T. K. (1998). “The Random Subspace Method for Constructing Decision Forests”. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 20(8), 832–844.
- Louppe, G. and Geurts, P. (2012). “Ensembles on Random Patches”. *Machine Learning and Knowledge Discovery in Databases* (ECML PKDD 2012), LNCS 7523, Springer, 346–361.
- scikit-learn developers. User Guide, “Bagging meta-estimator”. [scikit-learn.org, modules/ensemble](http://scikit-learn.org/modules/ensemble).

9. Key terms

Term	Meaning
Observation	One record of the data: one row of the data table
Feature	An input variable: one column of the data table
Target	The output we predict

Term	Meaning
Bootstrapping	Drawing random samples of the data, with replacement, to train each base model
Unstable model	A model that changes a lot when the training data changes a little, such as a fully grown tree
Aggregation	Combining the base models' predictions into one: mode for classes, mean for numbers
With replacement	Sampling in which each drawn item is put back, so it can be drawn again
Out-of-bag observations	The observations a base model never saw because its bootstrap sample missed them (about 37%)
Pasting	Bagging with observations sampled without replacement
Random subspaces	Bagging in which each model gets all observations but a random subset of features
Random patches	Bagging in which each model gets random observations and random features